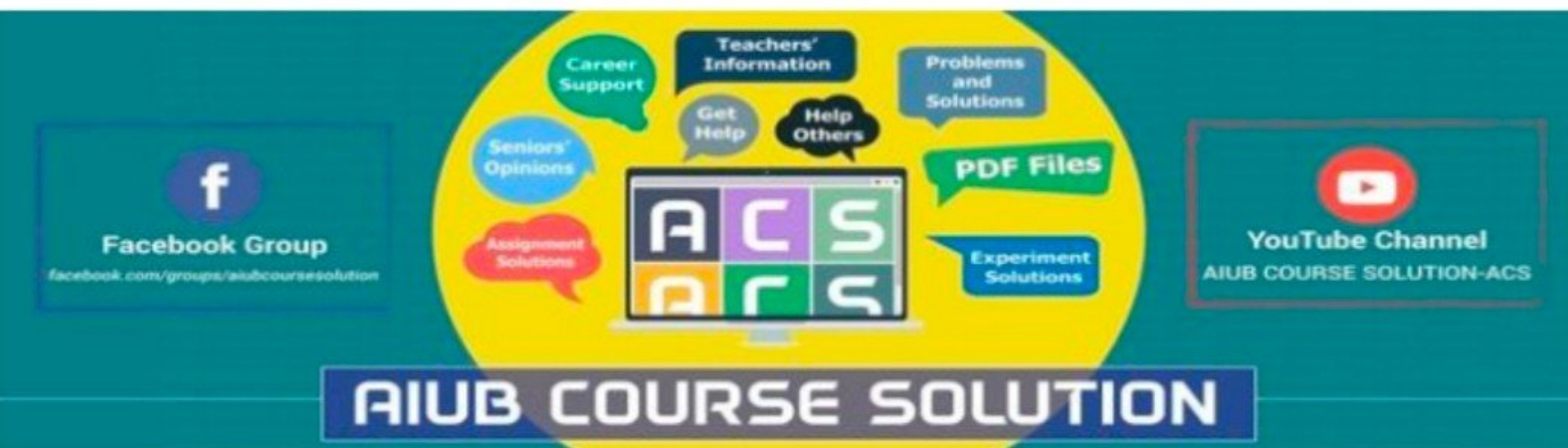




TOGETHER WE CAN ACHIVE MORE

COURSE NAME: Object Oriented Programming 1
SEMESTER: Summer 24-25 (Lecture 1)

SOLVED BY

Awishi Ghosh

15/07/25

Java class (Theory)

★ Bold line for MCQ → 25

In java we will call function as method.

Architecture → কিছু আসবে না

primitive → pre-defined →

Non-primitive → ~~চাংগেবল~~ Changable → User defined

class → ~~নিচক~~ Non-primitive

★ Bit, Bytes → MCQ → Conversion

Data Type → Slide 6

↕
default value → There is a chart which is important.

Slide 9 → wrapper class → Important

wrapper class এর through তে primitive change করা সম্ভব।

Implicit Type casting → ছোট থেকে বড় (int → float)

Same size হলে Casting possible না

All Bold lines of Slide 1 $\Rightarrow$

* Java was conceived by James Gosling,

* was initially called Oak but later renamed Java.

* primary motivation $\rightarrow$ platform independent.

↳ not the internet.

* They all have state and behavior.

* Object oriented programming (OOP) is programming methodology that helps organize complex programs using inheritance, encapsulation, and polymorphism.

* process-oriented model was earlier approaches to object oriented programming.

* Two element of programs $\rightarrow$ code and data

□ process-oriented model $\Rightarrow$

- Code acting on data
- functions. [small parts called]
- Follows top down approach

• Based on unreal world.

* Importance of oop =>

- Characterized by data controlling
- access to code
- number of entities called objects
- Follows bottom up approach
- increases code reusability and maintainability

Java (as oop language)

impossible to code without class and object.

Supports encapsulation, abstraction,

inheritance, polymorphism.

Data type & Type casting

primitive Data Types $\rightarrow$ boolean, byte, short, int, long
char, float, double.

Non-primitive Data Types $\rightarrow$ String, System, Scanner

Data Type Memory Size

boolean $\rightarrow$ 1 byte

byte $\rightarrow$ 1 byte

short $\rightarrow$ 2 bytes

int $\rightarrow$ 4 bytes

long $\rightarrow$ 8 bytes

char $\rightarrow$ 2 bytes

float $\rightarrow$ 4 bytes

double $\rightarrow$ 8 bytes

1 byte = 8 bits

4 bytes = 32 bits

2 bytes = 16 bits

Important
for
MCQ

Boolean $\rightarrow$ No Minimum and Maximum Value
Default Value False

byte $\rightarrow$ min value = -2^7
max = $2^7 - 1$
default = 0

short $\rightarrow$ min = -2^{17}
max = $2^{17} - 1$ * Formula ;
default = 0 if bits = n

int $\rightarrow$ min = -2^{31}
max = $2^{31} - 1$ min value = -2^{n-1}
default = 0 max value = $2^{n-1} - 1$

long $\rightarrow$ min = -2^{63}
max = $2^{63} - 1$
default = 0L

char $\rightarrow$ min = -2^7 '\U0000'
max = $2^7 - 1$ '\UFFFF'
default = '\U0000'

* value of sign bit is 1 $\rightarrow$ negative number

* value of sign bit is 0 $\rightarrow$ positive number

• A wrapper class is a Java library class whose object wraps or contains a primitive data type.

Compile

javc hello.java

In Java we use [+] for combining.

access → इकाई किछु रिजल्ट त्रिपु किछु Use करी।

Next class, slide 8

Slide 2.2 ⇒

- templates or blueprints for objects [classes].
- Object is an instance of a class.
- process of creating an object is called instantiation.
- Attributes of an object → instance variables
- The methods of an object → instance methods.
-

22-07-25

Java

Quiz 1 - 29-07-25
(Tuesday)

Mark distribution $\Rightarrow$

MCQ $\rightarrow$ 20

Output $\rightarrow$ 5

OBE - 10

Code writing - 15

Attendance - 10

Quiz 1 out of 2 (Best one) $\rightarrow$ 30

lab task $\rightarrow$ 10

*Inheritance w₃

Important $\rightarrow$ Array, Array of Object

Quiz 1 $\rightarrow$ 1.1 - 3.1

pattern $\rightarrow$ MCQ + Code (lab or code)
writing

class → without creating any object
Variable - access করতে পারবে।

overloading → function same and parameter
ভিন্ন।

Static block → যদি for exam.

*In Java we use String as Object.

Immutable → Value of String Change করা
যায় না।

2nd. ব্যাক্তকে দায়া close অঙ্ক → Scope

class {

data members

+

methods (behave)

}

→ Encapsulation

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PAGE: <https://www.facebook.com/acsaib>



<https://www.youtube.com/channel/UCC3KjA8kstFtM-2CxVr-jcg>



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